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REVIEW ARTICLE

Cardioprotection: Where to from here?

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Abstract The size of the myocardial infarction remains an important therapeutic target, because heart attack size correlates with mortality and heart failure. In this era, myocardial infarct size is reduced primarily by timely reperfusion of the infarct related coronary artery. Whereas numerous pre-clinical studies have shown that certain pharmacologic agents and therapeutic maneuvers reduce myocardial infarction size greater than reperfusion alone, very few of these therapies have translated to successful clinical trials or standard clinical use. In this review we discuss both the recent successes as well as recent disappointments, and describe some of the newer potential therapies from the preclinical literature that have not yet been tested in clinical trials.

Keywords Acute myocardial infarction · ST elevation myocardial infarction · Myocardial infarct size reduction · Beta blockade · Adenosine · Hypothermia

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Background

In the United States, there are ~15.5 million people with coronary artery disease. There are about 750,000 myocardial infarctions every year: 550,000 are new and 200,000 are recurrent. Approximately every 42 s someone in the United States has a myocardial infarction. Approximately 34 % of people who experience a coronary artery event in a given year, die of it and about 15 % who experience a myocardial infarction will die of it. In-hospital death rates for myocardial infarctions have dropped since the advent of reperfusion therapy; however they still can run 6–10 % and 30-day mortality rates hover at about 10 %. A major concern is that despite contemporary therapies for acute myocardial infarction, about one quarter of all myocardial infarction patients will experience heart failure at one year [1]. This problem is not confined to the US, but it is global. According to the World Health Organization, there are ~17.3 million deaths /year due to cardiovascular disease worldwide; 7,254,000 are due to ischemic heart disease.

The size of the myocardial infarction remains an important therapeutic target. Myocardial infarct size correlates with mortality and heart failure. Even a 5 % reduction in the size of a heart attack improves clinical outcome [2, 3]. The major way that clinicians are reducing myocardial infarct size is by instituting timely reperfusion of the infarct related coronary artery. However, a sobering article in the New England Journal of Medicine suggested that although the time from ambulance arrival to re-opening the occluded coronary artery in a catheterization laboratory has shortened in the United States to about 90 min, further shortening of this time has not further reduced mortality [4]. This study suggested that additional therapies are needed to further improve outcomes, and this might involve the use of adjunctive agents, given with coronary reperfusion, to further reduce the size of the myocardial infarction. Whereas numerous pre-clinical studies have shown

that certain pharmacologic agents and therapeutic maneuvers reduce myocardial infarction size more than reperfusion alone, very few of these therapies have translated to successful clinical trials or standard clinical use. We and others have previously published papers dealing with this discrepancy. In this personal perspective, celebrating the 20th anniversary of *Cardiovascular Drugs and Therapy*, we will discuss our observations and opinions including both the recent disappointments as well as recent successes and describe some of the newer potential therapies from the preclinical literature that have not yet made it into clinical trials.

We point out that there have been several recent, well-written and more systematic reviews, narrative reviews and consensus statements on this topic that the reader might find useful [5–11].

Disappointments

From 2000 to 2012 there were a number of pharmacologic agents that initially looked promising in experimental animal studies but simply did not reduce infarct size or improve clinical outcome when tested in clinical trials. These include: 1) CD11/CD18 leukocyte integrin receptor inhibitor, 2) the sodium/hydrogen exchange inhibitor, eniporide, 3) caldaret, an intracellular calcium handling modulator, 4) Nicorandil, the potassium channel opener-nitrate like vasodilator, 5) Glucose-insulin-potassium, 6) delcasertib, the delta protein kinase C inhibitor, and 7) erythropoietin, as reviewed by Gerczuk PZ and Kloner RA [12].

Since 2012 to 2016 there are other therapies that appeared promising, sometimes in small clinical trials that were not effective when tested in larger multicenter studies. Examples are postconditioning [13–15], chemical postconditioning by inhibition of the mitochondrial permeability pore with cyclosporine A [16, 17] and mitochondrial targeted peptides. The recent DANAMI 3-iPOST trial showed that ischemic postconditioning with brief episodes of balloon occlusion immediately following reperfusion for STEMI was not superior to conventional primary percutaneous coronary intervention. Post-conditioning did not improve the primary outcome (composite of all-cause mortality, hospitalization for congestive heart failure, 10.5 %) vs. conventional percutaneous coronary intervention (11.2 %, $p = 0.66$) [18]. A mitochondrial targeted protective agent, MTP-131 was promising in some experimental models, although to a variable degree and failed to significantly reduce myocardial infarct size in a clinical trial [19]. In our opinion it is unlikely that the postconditioning pathway or mitochondrial permeability pore blockers will continue to be studied in clinical trials of ST elevation myocardial infarction (STEMI). A recent review discusses some potential reasons for discrepancies between preclinical and clinical trials of conditioning [20].

Potential Successes

Adenosine

Despite the failures of the above clinical trials and a general sense in the academic cardiology community that adjunctive therapies simply do not work, there have been some bright spots in the literature. In 2 separate clinical trials AMISTAD 1 and AMISTAD 2 [21, 22], high dose adenosine infusions significantly reduced anterior wall myocardial infarct size. In the AMISTAD 2 trial high-dose adenosine reduced the median infarct size from 27 % of the left ventricle to 11 %, a substantial 59 % reduction in infarct size. Although the major clinical endpoints of death and heart failure were not significantly reduced by adenosine in this study, a subgroup analysis of patients reperfused within less than 3.2 h of symptom onset (median time to reperfusion) showed that adenosine reduced the composite endpoint of death and congestive heart failure from 17.2 % in the placebo group to 12.1 % in the pooled adenosine group ($p = 0.023$) [23]. However, patients reperfused after 3.2 h of symptoms did not show any improvement in the composite clinical outcome with adenosine versus placebo. Not all studies with adenosine have been positive. Studies in which the adenosine was administered by an intracoronary bolus or low dose infusions have shown mixed results [24–27]. That finding is not so surprising, given that adenosine's half-life in the blood is very short (seconds). Of note, however, a recent meta-analysis showed less heart failure and less no-reflow in patients receiving intracoronary adenosine as an adjunct to reperfusion [28]. The mechanism by which adenosine worked is not known but some of the theories include coronary vasodilation, decrease oxygen demand by afterload and heart rate reduction, triggering a preconditioning second messenger pathway, or an antiplatelet effect. Because 2 sizable studies showed that adenosine reduced acute anterior wall STEMI and improved clinical outcome in patients reperfused in a timely manner, adenosine should continue to be pursued as a cost effective therapy to be used with reperfusion. However, given that it is now generic, it is unlikely any large scale studies will be performed.

Remote Ischemic Conditioning

One very exciting area that shows promise is the concept or remote ischemic conditioning. In 1993, Przyklenk in our laboratory was the first to show that brief episodes of ischemia in one vascular bed could protect another vascular bed from ischemic damage [29]. She observed that preconditioning the circumflex coronary artery bed with brief episodes of ischemia/reperfusion reduced infarct size when the left anterior descending coronary artery was occluded for 60 min followed by reperfusion. Gho et al. observed that a 15-min anterior mesenteric artery occlusion and reperfusion was

equally protective in reducing experimental infarct size as ischemic preconditioning of the infarct-related artery [30]. This study showed that ischemia plus reperfusion in a remote organ protected the myocardium. However, the clinical application of mesenteric ischemia would be difficult. If a limb could be made ischemic (for example by applying a blood pressure cuff on the arm or leg) and protect the myocardium, then such a therapy could be clinically relevant. Subsequent work by Birnbaum in our laboratory showed that lower limb ischemia induced by a partial occlusion of the femoral artery and electrical stimulation of the gastrocnemius muscle in the anesthetized rabbit, prior to a proximal coronary artery occlusion could significantly reduce myocardial infarct size [31]. There have been numerous studies verifying that remote ischemic conditioning could benefit organ ischemia and reperfusion. The mechanism may involve a protective humoral substance produced by the ischemic limb and then transferred to the infarcting organ or perhaps neurologic reflexes or a combination of both mechanisms. Botker studied remote ischemic conditioning before hospital admission as a complement to angioplasty [32]. Three hundred thirty three patients with a first myocardial infarction were randomized to either primary percutaneous intervention with or without remote ischemic conditioning (4 cycles of 5-min brachial artery cuff inflation and 5 min cuff deflation). The median salvage index determined by myocardial perfusion imaging was 0.75 in the remote conditioning group versus 0.55 in the control group ($p = 0.03$). In addition a regression analysis revealed that for any given area at risk, there was less infarction in the remote conditioning group. In a follow-up study, reported by Sloth et al., at 3.8 years, patients who had received the remote conditioning, had a significant reduction in major adverse cardiac and cerebrovascular events, which included the composite of all-cause mortality, myocardial infarction, readmission for heart failure, ischemic stroke, and transient ischemic attack. All-cause mortality was also lower in the remote conditioning group [33]. However, as this study was a follow-up of a study initially designed to measure infarct size as the primary endpoint, it might have been underpowered for the purpose of testing clinical benefit, and this is a potential limitation of the study. Recent papers by Yellon et al. [34], Prunier et al. [35], and Crimi et al. [36], have shown that remote ischemic conditioning reduced enzymatic markers of acute myocardial infarction as reviewed by Sharma et al. [37]. Yamanaka et al. found that remote conditioning decreased the incidence of contrast induced acute kidney injury [38]. Two meta-analyses, one by Zuo [39] and one by Yetgin et al. [40] showed that remote ischemic conditioning had benefits in acute coronary syndromes. A meta-analysis by Le Page et al. [41] showed that remote ischemic conditioning in the setting of primary PCI for myocardial infarction, reduced CK-MB peak and area under the curve and reduced troponin peak and troponin area under the curve. In addition remote conditioning reduced both long term major adverse cardiovascular events as well as all-cause mortality. Remote ischemic conditioning has not been as consistent when given prior to cardiac surgery to try to limit damage. There, the data are much more mixed. Two recent studies published in the *New England Journal of Medicine* failed to show a benefit of remote ischemic conditioning in patients undergoing cardiac surgery [42, 43]. At present, remote ischemic conditioning remains a very cost effective, low tech way to reduce myocardial infarct size in patients with STEMI, and we hope that it is adopted by various guideline committees.

Therapeutic Hypothermia

Therapeutic hypothermia is one of the most powerful techniques to reduce myocardial infarct size [44–46]. When done correctly it can markedly reduce the amount of necrosis that occurs with myocardial ischemia/reperfusion. In a recent series of studies we observed that bringing the core temperature down from 37 or 38 °C to <32 °C in rabbit and rat models could be achieved within a few minutes using a new convection-immersion device called the ThermoSuit system. The size of the myocardial infarctions were markedly reduced in hypothermic hearts by 70–80 %, and within the ischemic risk areas, the zones of no-reflow were reduced [46]. Inducing hypothermia at the time of cardiac catheterization for re-establishing coronary patency in the setting of an STEMI, has been problematic. Endovascular cooling catheters have not always cooled the patient quick enough. Induction of hypothermia can interrupt the flow in the catheterization laboratory, where opening the occluded infarct coronary artery is a paramount goal. In early studies, overall results with therapeutic hypothermia induced with endovascular catheters were negative, but subgroups of patients who achieved early hypothermia (< 35 °C) plus early reperfusion in the setting of anterior wall MIs had smaller infarcts than patients in normothermic groups. In a study by Gotberg et al., rapid systemic cooling to <35 °C using a combination of cold saline plus endovascular cooling reduced the size of myocardial infarcts by 38 % at day 4 ($p = 0.041$) [47]. Erlinge et al. studied a combination of cold saline plus endovascular cooling in patients with STEMI. While the overall results were negative, a subgroup of patients presenting within four hours who had anterior STEMI did show a reduction of infarct size with cooling [48]. The same author pooled patients from the RAPID-MI-ICE and CHILL-MI studies [49]. Patients who reached a temperature of 34.7 degrees before reperfusion demonstrated a reduction in infarct size (Relative Reduction (RR) = 15 %, $p = 0.046$). The greatest infarct size reduction occurred in anterior infarcts that were reperfused early (RR = 31 %, $p = 0.01$). Patients with risk zones of >30 % of the left ventricle had significant decreases in infarct size with hypothermia (40.5 versus 55.1 % of the ischemic risk zone,

$p = 0.03$), whereas those with risk zones <30 % of the left ventricle had less reduction of infarct size. Finally, hypothermia also reduced heart failure following STEMI. A study in which hypothermia was induced by intraperitoneal cooling was negative [50]. However, here there was an increase in adverse events in patients receiving peritoneal hypothermia, including 3 cases of stent thrombosis, not seen in the control group. In addition there was some concern about patient comfort, volume overload, inhibition of breathing by volume impinging upon the diaphragm and the invasiveness of peritoneal catheters and volume infusion during an already stressful setting of STEMI. Clearly newer techniques for inducing hypothermia quickly that are safe, comfortable and efficacious are needed. There are ongoing studies using different catheter techniques, perfusion of cold fluids directly into the coronary circulation or retrogradely through the coronary sinus.

Beta Blockade

Experimental studies carried out in the 70s and 80s showed that beta blockers administered during ischemia, protected the myocardium, reduced myocardial infarct size and preserved intracellular organelles [51–53]. Clinical trials carried out in the pre-reperfusion and thrombolytic eras that attempted to show that beta blockers could reduce infarct size and improve clinical outcome were mixed. Recently, Ibanez et al. reported the results of the METOCARD-CNIC (Effect of Metoprolol in Cardioprotection During an Acute Myocardial Infarction) trial [54]. Patients with acute anterior wall myocardial infarctions were randomized to intravenous metoprolol (15 mg IV bolus) or saline prior to reperfusion therapy. Myocardial infarct size was measured using cardiac magnetic resonance imaging days 5–7 after admission. The patients that received early intravenous metoprolol demonstrated smaller myocardial infarctions (25.6 g) compared to the control group (32.0 g, $p = 0.012$) and also demonstrated a better left ventricular ejection fraction. The improved ejection fraction in the metoprolol group persisted at 6 months and was associated with less hospitalization for heart failure. The same investigators recently reported that the longer metoprolol was on board prior to reperfusion, the smaller was the myocardial infarct size [55], findings which mirror preclinical studies [56]. However, another very recent study by Roolvink el al. involving some of the same coauthors surprisingly did not show a reduction in myocardial infarction when intravenous beta blocker was administered to patients with acute myocardial infarction [57]. This study was a double-blind, placebo controlled, multicenter study of 683 patients with ST segment elevation myocardial infarction randomized to receive two 5 mg boluses of IV metoprolol; the first bolus was given in the ambulance; the second was given in the catheterization laboratory shortly before reperfusion. The primary endpoint was infarct size determined by cardiac magnetic resonance imaging at 30 days, which did not differ between the placebo group (15 % of the LV) versus the metoprolol group (15 %). The incidence of malignant ventricular arrhythmias was lower in the metoprolol group. Why was there a difference in the outcomes for infarct size between the METOCARD-CNIC trial and Roolvink (also known as the EARLY-BAMI trial) study? There are several potential explanations: 1) The Roolvink study included all ST elevation myocardial infarcts while the METOCARD-STUDY included only anterior wall MIs, resulting in smaller infarct sizes in general in the Roolvink study, which is problematic since cardioprotective agents are most likely to show benefit in larger MIs; 2) In the Roolvink study the dose of beta blocker during most of the ischemic phase (5 mg) may have been too low, since the second dose was not received until on average 14 min prior to reperfusion and there was no difference in blood pressure prior to the second dose between the 2 groups. In other words, the patients may not have been adequately beta blocked during the crucial period of ischemia; 3) 19 % of the patients in the Roolvink study were already on long term beta blocker (an exclusion criterion in the METOCARD-CNIC study) which could have resulted in smaller infarcts in the control group; 4) differences in the timing of when the cardiac magnetic resonance imaging was performed and the demographics of those patients who underwent this imaging technique; 5) differences due to blinding and use of placebo. Additional discussion of the differences between the METOCARD-CNIC and EARLY-BAMI trial can be found in a recent editorial [58]. This most recent study suggests that the issue of the usefulness of early intravenous beta blockers for limiting infarct size in patients is not yet resolved; future studies should consider higher doses of metoprolol or other beta blockers, given as soon as possible after the diagnosis of MI, careful assessment of the degree of beta blockade, and should be limited to anterior (large) MIs and those patients not already on chronic beta blockers. Adequately sized and carefully controlled Phase 3 clinical trials are needed. Whereas some trials could be supported by pharmaceutical companies, others (such as beta blockers in anterior wall STEMI) would require government, such as NIH, funding.

Other Promising Therapies

There are some other therapies that appear promising for reducing myocardial infarction in the clinical setting. We will mention some of them here, but caution that some of these are based on only one or two clinical trials and will likely need to be repeated in larger or multiple trials before they are adopted into clinical use. One is the concept of delivering supersaturated oxygen as was tested in the AMIHOT 1 and 2 trials. A combined analysis of these 2 trials suggested that hyperoxemia can reduce myocardial infarct size from 25 % in controls to 18.5 % in treated patients [59]. Kitakaze showed

in the J-WIND trial that atrial natriuretic peptide reduced myocardial infarct size in patients by a modest amount (14.7 %) and also improved left ventricular ejection fraction [60]. Exenatide is an incretin mimetic indicated for the treatment of type 2 diabetes mellitus. Lonborg et al. showed that exanatide reduced myocardial infarct size from 13 g of tissue to 9 g, $p = 0.008$ when given as an intravenous infusion prior to primary percutaneous intervention, especially in patients with short ischemic time from onset of symptoms (<132 min; $p = 0.006$) [61, 62].

Preclinical Experimental Studies of Promising New Pharmacologic Agents for Reducing Myocardial Infarct Size not yet Tested in Clinical Trials

Despite the fact that hundreds of potential cardioprotective adjunctive therapies have been reported in animal studies during the last several decades, many have failed in clinical trials [5, 63]. No effective therapy for reducing myocardial reperfusion injury in STEMI patients is available so far [64]. There remains a need to continuously search for effective pharmacologic agents to reduce both ischemic and reperfusion induced myocardial damage. Numerous pharmacologic agents that can reduce infarct size in various experimental models and in different species have been reviewed in detail elsewhere [65, 66]. This following discussion reviews promising experimental reports published in the past year.

Mitochondria are a major player in cardiac ischemia-reperfusion injury, because mitochondria constitute 30 % of the total myocardial cell volume and are the major source of ATP for the cell as well as a significant source of reactive oxygen species. Mitochondrial structure and function are severely altered during ischemia and after restoration of blood flow to the ischemic myocardium. Although attempts at targeting mitochondrial permeability transition pore [67], mitochondrial reactive oxygen species production [68], and cardiolipin [69] have been clinically disappointing, mitochondrial quality control mechanisms remain to be a therapeutic target in protection against cardiac ischemia-reperfusion injury. Zhang et al. [70] administered cloxyquin (5-chloroquinolin-8-ol), an agent thought to be mitochondrial-protective, to mice by tail-vein injection. Twenty minutes later, left coronary artery was occluded for 30 min followed by 24 h of reperfusion. Cloxyquin treatment produced a significant reduction of infarct size compared to the control group. The mechanism for cloxyquin-induced cardioprotection was uncoupling mitochondria and simultaneously inducing autophagy. Another agent, diosgenin [71], has demonstrated cardioprotective effects against myocardial reperfusion injury in an isolated buffer-perfused rat heart model, in which global heart ischemia was induced by 30 min occlusion of aortic flow followed by 90 min reperfusion. The study has demonstrated that diosgenin activated mitochondrial ATP-sensitive potassium channels in the mitochondrial inner membrane and suppressed mitochondrial calcium overload and overproduction of oxygen derived free radicals during simulated ischemia/reperfusion. Therefore, this mitochondrial KATP channel opener might modulate mitochondrial function during myocardial ischemia/reperfusion. The cardioprotective effects of diosgenin need further investigations in in vivo animal myocardial ischemia/reperfusion models before clinical trials are considered.

Hydrogen sulfide (H_2S) is an endogenously produced transmitter that regulates cardiovascular function, and has emerged as an important cardioprotective molecule with potential for clinical applications. Meng et al. treated rats with morpholin-4-ium-methoxyphenyl-morpholino-phosphorodithioate (GYY4137) by intraperitoneal injection for 7 days. GYY4137 slowly releases low but consistent concentrations of H_2S over several hours to mimic the time course of H_2S release in vivo. The pretreated rats were then subjected to 30 min of left coronary artery occlusion followed by reperfusion for 24 h. H_2S slow-releasing agent GYY4137 significantly reduced infarct size and improved cardiac function compared to vehicle-treatment [72]. Zhao et al. designed and synthesized thirty-three N-mercapto (N-SH)-based H_2S donors and tested the protective effects of two representative donors NSHD-1 and NSHD-2 in a murine model of myocardial ischemia reperfusion injury. The mice were subjected to 45 min of left coronary artery occlusion followed by 24 h reperfusion. At the time of reperfusion, NSHD-1, NSHD-2, or vehicle was injected directly into the left ventricular cavity. Both NSHD-1 and NSHD-2 significantly reduced infarct size compared to vehicle treatment. The authors suggested that N-mercapto species are potent H_2S donors and that they possess potential therapeutic benefits for myocardial ischemia/reperfusion injury [73].

Compared to nondiabetic patients, diabetics are more susceptible to damage during cardiac stress, such as impaired calcium homeostasis, mitochondrial dysfunction, and increased oxidative stress [74]. So, it seems reasonable to consider that an increase in insulin sensitivity might be cardioprotective. Metformin is a biguanide used for the treatment of type 2 diabetes mellitus, and exerts its protective effect against ischemia/reperfusion injury through several mechanisms. For example, it can activate phosphatidylinositol-3-kinase and Akt, kinases belonging to the reperfusion injury salvage kinase (RISK) pathway [75]. Oidor-Chan et al. [76] induced diabetes by a single streptozotocin dose (70 mg/kg, i.p.) in the rat. Eight weeks later, the rats received 14 days treatment with metformin versus saline, and then were subjected to left coronary artery occlusion for 30 min followed by 120 min of reperfusion. Metformin treatment significantly reduced infarct size compared to the control group. The results demonstrated that metformin may represent a novel treatment strategy to limit ischemia/reperfusion injury in diabetic patients with myocardial infarction. Although in the recent GIPS-III

randomized clinical trial, metformin as adjunct to primary percutaneous coronary intervention was not associated with short-term improved left ventricular function after acute myocardial infarction in patients without diabetes [77], future clinical studies are needed to help clarify whether metformin has a cardioprotective effect against ischemic myocardial injury in patients with diabetes.

The reasons that adjunctive therapies often appear positive in experimental animal studies but then fail to translate to a beneficial effect in clinical trials have been discussed in detail in several review papers [5, 63, 64, 78, 79]. In brief, one potential reason is that the myocardial infarct is induced in the normal coronary artery of a healthy and young animal in nearly all preclinical studies, while atherosclerosis is common in the aged STEMI patients. Moreover, some patients usually have co-morbidities and concomitant medications. For example, some STEMI patients received platelet inhibitors, such as ticagrelor, which may have cardioprotective effects and might mask the benefit of other tested therapeutic strategies [80]. Another major factor is the mechanisms underlying myocardial reperfusion injury are complex and remain incompletely understood.

Conclusion

Whereas many clinical studies of adjunctive therapy have failed to reduce myocardial infarct size above and beyond reperfusion alone, some studies show promise including intravenous adenosine infusions, remote ischemic conditioning, therapeutic hypothermia, beta blockade and others (Table 1). Since there still is an unmet need to improve outcome after STEMI, future attempts to better reduce the size of myocardial infarction should continue [81].

Where do we go from here? Better techniques are needed to induce hypothermia quickly and safely prior to reperfusion. Large, Phase 3 trials that are adequately sized and controlled should be considered for those interventions that have already shown promise in limiting infarct size in clinical trials. These therapies include early administration of beta blockers, remote ischemic conditioning, and intravenous (but probably not intracoronary) adenosine infusion. Infarct size should be measured by both visual (magnetic resonance imaging) and enzymatic techniques. Major adverse cardiovascular events and long-term assessment of LV remodeling and function should be considered.

Table 1 Agents for reducing infarct size

Agents that have demonstrated reduction of myocardial infarct size in some clinical trials
1 Remote ischemic conditioning
2 IV adenosine infusion
3 Therapeutic hypothermia (initiated prior to reperfusion in large anterior MIs)
4 Early beta blockade
5 Others: supersaturated oxygen, atrial natriuretic peptide, exenatide
Agents that have failed to reduce infarct size in clinical trials
1 CD11/CD18 leukocyte integrin inhibitor
2 Eniporide (Na/H exchange inhibitor)
3 Caldaret (intracellular calcium-handling moderator)
4 Nicorandil
5 Glucose/insulin/potassium
6 Delcasertib (delta-protein kinase C inhibitor)
7 Post conditioning
8 Mitochondrial protective agents (cyclosporine, MTP-131)

Compliance with Ethical Standards

Disclosure of Potential Conflict of Interest RAK's laboratory receives grant support from Stealth Biotherapeutics, Faraday and Servier to study experimental agents in myocardial infarct models. RAK has been a consultant to IC Therapeutics/Endothelix.

Other authors have no potential conflict of interest.

Ethical Approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed for studies performed in RAK's Laboratories.

Human and Animal Rights This article does not contain any studies with human participants performed by any of the authors.

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